



Computer Networks 06 | Problem Solving on Subnetting

Part-2 | CS & IT | GATE Crash Course

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Summary

AI Notes

AI Slides

Text Notes

Screenshots

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Q.9 Consider the class-c Network with 7 subnets and 25 Host per subnet. What will be the subnet mask for this network

Best

A $8 + 8 + 8 + 3 = 27 \rightarrow 1's$
255.255.255.224 [128+64+32] class-C

B $8 + 8 + 8 + 3 = 27 \rightarrow 1's$
255.255.255.44 [32+8+4] $\frac{NID}{24} \frac{H}{2}$

C $8 + 8 + 8 + 6 = 30 \rightarrow 1's$
255.255.255.63 [00111111]

D $8 + 8 + 8 + 3 = 27 \rightarrow 1's$
255.255.255.112 [01110000] $\frac{24}{NID}$

MSQ

No. of 1's in the
SM = NID + SID = 24 + 3 = 27

No. of 0's in the
S.M = HID = 5 bit

11111111.11111111.11111111.00000000
255.255.255.224

ABD

MEWALLAH

▶ 1:16:23

Q.10

Consider the class-B Network with 180 subnets and 200 Host per subnet. What will be the subnet mask for this network



$192:11000000$

$$180 \times 200 \leq 2^{16} - 2$$

MSQ

$$36,000 \leq 65,534$$

class-B

No. of 1's in the

S.M = NID + SID

$$= 16 + 8 = 24$$

No. of 0's in

the S.M = HID

$$= 8$$

$$8 + 8 + 8 = 24 \rightarrow 1's$$

255.255.255.0

$$8 + 8 + 4 + 4 = 24 \rightarrow 1's$$

255.255.240.240

$$8 + 8 + 6 + 3 = 25 \rightarrow 1's$$

255.255.252.112

$$8 + 8 + 6 + 2 = 24 \rightarrow 1's$$

255.255.252.192

$$240: 11110000$$

$$112: 01110000$$

$$252: 11111100$$

$$\frac{NID}{16} \quad \frac{HID}{16}$$

180 subnets

$$\frac{16}{NID} \quad \frac{8}{SID} \quad \frac{8}{HID}$$

1:24:38

Q.11

If Direct Broadcast address is 200.200.200.31 Which of the following can be subnet mask ?



MSG

A 255.255.255.192

B 255.255.255.224

C 255.255.255.248

D 255.255.255.128

$$200.200.200.00011111$$

NID

HID can be maximum

5 bit

HID = 5 bit

1:32:21

20:40

Q 255.255.255.192

255.255.255.11000000

NID SID HID

192 64

SID HID

00 1 1 1 1 1 1 → 63 ✓

01 1 1 1 1 1 1 → 127 ✓

10 1 1 1 1 1 1 → 191 ✓

11 1 1 1 1 1 1 → 255 ✓

▶ 1:37:12

20:43

Q 255.255.255.192

255.255.255.11000000

NID SID HID

192 64

SID HID

00 1 1 1 1 1 1 → 63 ✓

01 1 1 1 1 1 1 → 127 ✓

10 1 1 1 1 1 1 → 191 ✓

11 1 1 1 1 1 1 → 255 ✓

GATE WALLAH

Q 255.255.255.11000000

NID SID HID

SID HID

000 1 1 1 1 1 1 → 31 ✓

001 1 1 1 1 1 1 → 63 ✓

010

011

100

101

110

111

▶ 1:40:11

$$\textcircled{a} \quad \begin{array}{c|c|c} 255 & 255 & 255 \\ \hline \text{NID} & \text{SID} & \text{HID} \end{array} \quad \begin{array}{c|c} 1111 & 000 \\ \hline \text{SID} & \text{HID} \end{array}$$

$$00011 \ 111 \rightarrow 31$$

$$31: \underline{00011111}$$

AD Rule

$$\text{HID} = 3 \text{ bit (HID} \leq 5 \text{ bit)}$$

$$\textcircled{X} \quad \begin{array}{c|c|c} 255 & 255 & 255 \\ \hline \text{NID} & \text{SID} & \text{HID} \end{array} \quad \begin{array}{c|c} 10000000 \\ \hline \text{HID} \end{array}$$

AD Rule

$$\begin{array}{c|c} \text{SID} & \text{HID} \end{array}$$

$$\text{HID} = 7 \text{ bit (HID} \leq 5 \text{ bit)}$$

No

$$\begin{array}{c|c} 0 & 1111111 \rightarrow 127 \\ 1 & 1111111 \rightarrow 255 \end{array}$$

▶ 1:46:35

Q.12

If Direct Broadcast address is 200.200.200.31 Which of the following can be subnet mask ?



A

255.255.255.192

B

255.255.255.224

C

255.255.255.248

D

255.255.255.240

$$\textcircled{a} \quad \begin{array}{c|c|c} 255 & 255 & 255 \\ \hline \text{NID} & \text{SID} & \text{HID} \end{array} \quad \begin{array}{c|c} 11000000 \\ \hline \text{HID} \end{array}$$

$$\text{HID} = 6 \text{ bit (HID} \leq 5 \text{ bit)}$$

No

$$\begin{array}{c|c} 24 & \text{SID} \\ \hline \text{NID} & \end{array}$$

$$\begin{array}{c|c|c} 200 & 200 & 200 \\ \hline \text{NID} & \text{SID} & \text{HID} \end{array} \quad \begin{array}{c|c} 000 & 11 \\ \hline \text{HID} & \end{array}$$

GATE WALLAH

▶ 1:49:51

Q.12

If Direct Broadcast address is 200.200.200.31 Which of the following can be subnet mask ?



A

255.255.255.192

B

255.255.255.24

C

255.255.255.248

D

$$\textcircled{a} \begin{array}{c} 255 \cdot 255 \cdot 255 \cdot 11000000 \\ \hline \text{NID} \quad \text{SID} \quad \text{HID} \end{array}$$

HID = 6 bit (HID ≤ 5 bit)
No

OR

$$\begin{array}{c} \frac{24}{\text{NID}} \quad \frac{\text{SID}}{\text{---}} \quad \frac{\text{HID}}{\text{---}} \\ \text{---} \quad \underline{111111} \end{array}$$

$$31: 000 \underline{11111}$$

$$00 \underline{11111}$$

HID can be maximum 5 bit (HID ≤ 5 bit)

▶ 1:50:16

$$\textcircled{b} \begin{array}{c} 255 \cdot 255 \cdot 255 \cdot 11100000 \\ \hline \text{NID} \quad \text{SID} \quad \text{HID} \end{array}$$

HID = 5 bit (HID ≤ 5) ✓

OR

$$\begin{array}{c} \frac{\text{NID}}{24} \quad \frac{\text{SID}}{\text{---}} \quad \frac{\text{HID}}{\text{---}} \\ \text{---} \quad \underline{11111} \end{array}$$

$$31: 000 \underline{11111}$$

20:55

▶ 1:52:22

$$\textcircled{P} \quad \begin{array}{c|c|c} 255 & 255 & 255 \\ \hline \text{NID} & \text{SID} & \text{HID} \end{array} \quad \begin{array}{c} 11100000 \\ \hline \end{array}$$

HID = 5 bit (HID ≤ 5) for

OR

$$\begin{array}{c|c|c} \text{NID} & \text{SID} & \text{HID} \\ \hline 24 & \text{---} & \text{---} \\ \hline & \text{---} & \underline{11111} \end{array}$$

$$31: 000 \underline{11111}$$

$$\textcircled{P} \quad 255.255.255.248$$

$$\begin{array}{c|c|c} 255 & 255 & 255 \\ \hline \text{NID} & \text{SID} & \text{HID} \end{array} \quad \begin{array}{c} 11110000 \\ \hline \end{array}$$

HID = 3 bit (HID ≤ 3)

$$\textcircled{P} \quad 255.255.255.11110000$$

HID = 4 bit (HID ≤ 4)

▶ 1:54:31

Q.13

What could be the Network mask if DBA of a Network is 168.17.7.255?

MCQ

A

255.255.248.0

B

255.255.252.0

C

255.255.254.0

D

All the above

$$\begin{array}{c|c|c} 168 & 17 & 7 \\ \hline \text{NID} & \text{SID} & \text{HID} \end{array} \quad \begin{array}{c} 00000111 \\ \hline \end{array}$$

HID = 11 bit

can be maximum
= (HID ≤ 11 bit)

HID = 10 bit

HID = 9 bit

▶ 1:58:42

Q.14 A subnetted Class B network has the following broadcast address : 144.16.95.255. Its subnet mask

MCQ

(A) is necessarily 255.255.224.0 : $255 \cdot 255 \cdot 11100000 \cdot 00000000$ *HID = 13 bit*

(B) is necessarily 255.255.240.0 : $255 \cdot 255 \cdot 11110000 \cdot 00000000$ *HID = 12 bit*

(C) is necessarily 255.255.248.0 : $255 \cdot 255 \cdot 11111000 \cdot 00000000$ *HID = 11 bit*

(D) could be any one of 255.255.224.0, 255.255.240.0, 255.255.248.0

144.16.01011111.11111111
NID *HID can be maximum 13 bit (HID=13)*

▶ 2:03:06

Q.15 Given broadcast address 125.25.63.255 of a subnetwork. What can be Mask of this subnetwork :

MCQ

(A) 255.255.128.0 : $255 \cdot 255 \cdot 10000000 \cdot 00000000$ *HID = 15 bit*

(B) 255.255.192.0 : $255 \cdot 255 \cdot 11000000 \cdot 00000000$ *HID = 14 bit*

(C) 255.255.224.0

(D) Both B and C

DBA: 125.00011001.00111111.11111111
HID can be maximum 14 bit
HID = 14 bit

▶ 2:06:22

Doubt : 2:06:22 Question 15) DBA is of Class A, so max hid count should be HID <= 17 and not 14. but than first three option are right. please solve my doubt.

▶ 2:06:22

Very important-> Short cut for finding SM fails when continuous ones are not present ,\

Q.16 If Direct Broadcast address is 200.200.200.31
Which of the following can be subnet mask ?

A 255.255.255.192 : 255.255.255.11000000 $\text{HID} = 6\text{bit}$

B 255.255.255.224 : 255.255.255.11100000 $\text{HID} = 5\text{bit}$

C 255.255.255.198 : 255.255.255.11000110 $\text{HID} = 4\text{bit}$ (HIL)

D 255.255.255.128 : 255.255.255.10000000 $\text{HID} = 7\text{bit}$

200.200.200.000 11111
HID can be maximum 5 bit (HIL)